



ACQLERATE

MODULE 4

Data Analytics for Program Managers

*Use data to drive decisions, measure performance,
and communicate program health with confidence.*

4 Lessons · ~77 Minutes · Lesson Book Edition

Course Lessons

01	Program Metrics & KPIs for Defense PMs	13 min
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DATA ANALYTICS FOR PROGRAM MANAGERS · LESSON 1

Program Metrics & KPIs for Defense PMs

13 min

Identify and track the key performance indicators that matter most in defense program management.

Identify and track the key performance indicators that matter most in defense program management. Metrics won't fix a broken program, but they will at least keep you from being the last person in the building to find out it's broken.

Core Program Health Metrics

CATEGORY	METRIC	WHAT IT TELLS YOU	WARNING SIGN
Cost	CPI (EV/AC)	Cost efficiency of work performed	CPI < 0.90
Schedule	SPI (EV/PV)	Schedule efficiency vs. plan	SPI < 0.90
Cost	VAC (BAC - EAC)	Projected overrun/underrun at completion	VAC < -5%
Technical	TPM achievement %	Technical parameter vs. planned maturity	< 80% on schedule
Risk	Open high-risk items	Count of unmitigated high risks	> 5 high items
Quality	Defect closure rate	Rate of defect identification vs. resolution	Backlog growing
Schedule	Critical path float	Days of slack on program critical path	< 10 days float

- **Leading (predictive):** Staffing levels, test asset availability, GFE/GFI delivery status, risk burn-down rate, supplier delivery performance
- **Lagging (historical):** CPI, SPI, defect count, actual vs. planned milestones, budget execution rate
- **Best practice:** Track both: lead indicators give you time to act; lagging indicators confirm trends
- **Red flag:** If ALL your metrics are lagging, you're managing by looking in the rear-view mirror

Designing a Meaningful KPI Framework for Your Program

Mid-career PMs often inherit a metrics framework designed by someone else, or no framework at all (usually a spreadsheet named final_v3_ACTUAL.xlsx that nobody quite trusts anymore). Building a good one requires discipline: limit leading indicators to 5-7 (more creates noise), ensure each KPI has a data owner responsible for accuracy, and tie thresholds to contract performance criteria not arbitrary round numbers. The most important distinction: leading indicators (staffing ramp, BCWP/BCWS trend, open action item age) predict future

performance; lagging indicators (CPI, SPI, schedule slippage) confirm what already happened. Programs that only track lagging indicators are always reacting, never preventing. Build at least two leading indicators per major risk area. Review them weekly at the PM level, not just monthly at program reviews.

Metrics Manipulation: How Contractors Game KPIs and How to Catch It

Experienced senior PMs know that any metric a contractor is evaluated on will eventually be gamed, a rule that holds with roughly the reliability of gravity, yet somehow still gets treated as a fresh betrayal every time. The classic EVM manipulations: retroactive replanning (moving budget into periods where work was already done), schedule logic manipulation (adding constraints to remove negative float), and loading budget into early periods then underspending to generate positive CPI artificially. For technical metrics: systems that report “green” on readiness metrics by narrowing the denominator (counting only tested units, not all units required). The countermeasure: build cross-referencing requirements into your metrics framework. If CPI is 1.05 but cost-per-unit is rising, something is wrong. If schedule shows green but critical path items keep slipping, the schedule logic is suspect. Require contractors to explain anomalies: not just report metrics.

Key Terms

KPI

Key Performance Indicator: a measurable value demonstrating how effectively an objective is being achieved.

Technical Performance Measure (TPM)

A parameter that tracks achievement of key technical requirements against a planned profile.

Threshold

The minimum acceptable value for a performance parameter; program fails if not met.

Objective

The desired best value for a performance parameter; program tries to achieve this.

Leading Indicator

A metric that predicts future performance (e.g., staffing levels predict schedule performance).

Lagging Indicator

A metric that reports past performance (e.g., CPI shows what has already happened).

DATA ANALYTICS FOR PROGRAM MANAGERS · LESSON 2

Data-Driven Decisions & Visualization

12 min

Use charts, dashboards, and data storytelling to communicate program status and drive decisions.

Use charts, dashboards, and data storytelling to communicate program status and drive decisions. A dashboard that's all green and a program that's actually healthy are, unfortunately, two separate claims, and only one of them requires someone to update the data honestly.

The Core Chart Types Every PM Should Know

- **S-Curve:** Cumulative BCWS vs. BCWP vs. ACWP: the fundamental EVM visualization. Any divergence between the three lines tells the whole cost/schedule story.
- **Gantt/IMS Bar Chart:** Timeline view of all program activities, dependencies, and critical path. Updated monthly.
- **Risk Matrix (5×5):** Plot all risks by probability and impact. Shows at a glance where attention is needed.
- **Burn Rate Chart:** Monthly and cumulative obligation vs. plan. Reveals execution patterns.
- **TPM Trend Chart:** Actual technical parameter values vs. planned maturity profile over time.
- **Waterfall/Variance Chart:** Decompose cost variance by work package or contractor to identify root causes.

WHAT YOU WANT TO SHOW	BEST CHART TYPE
Overall cost & schedule health over time	S-Curve (BCWS/BCWP/ACWP)
Where cost variance is coming from	Waterfall chart by WBS element
Program schedule and dependencies	Gantt / IMS bar chart
Risk portfolio status	5×5 Risk matrix
Budget execution vs. plan by month	Bar chart (planned vs. actual burn)
Technical parameter trend vs. target	Line chart with threshold/objective bands
Comparing multiple programs	Radar/spider chart or normalized bar chart

Presenting Data to Non-Technical Decision Makers

Mid-career PMs spend a significant portion of their time briefing flag officers, SES officials, and congressional staff who have limited technical background. The most effective briefings for this audience: one chart per major message, direct headline titles that state the conclusion (not “CPI Trend” but “Cost Performance Is Recovering: CPI Up 0.08 in 90 Days”), and explicit “so what” statements. Avoid EVM jargon in flag-level briefings: translate: “negative SV” becomes “we are behind schedule”; “VAC” becomes “projected overrun at completion.” Nobody has ever walked out of a briefing wishing it had contained more acronyms. Spend 80% of your preparation on the narrative, 20% on the charts. Decision makers who can't understand your data can't support your program.

Building a Program Data Architecture: From Contract to Dashboard

Senior PMs on large programs should architect the data flow from the start: IPMR data → government EVMS analysis tool (SMART, Cobra, wInsight) → program office dashboard → leadership briefing. Each translation introduces error and delay; minimize handoffs. Define your data schema before contract award: what format does the contractor submit IPMR data in? What are your visualization requirements? For software programs, define your agile metrics pipeline: how does sprint velocity, defect backlog, and code coverage data flow from contractor dev tools (Jira, GitLab) to your dashboard? Programs that build data architecture retroactively spend 6-12 months cleaning inconsistent data before they can make it useful. Contract for the data architecture: not just the deliverable.

Key Terms

S-Curve

A cumulative cost/schedule plot showing planned vs. actual expenditure over time: the standard EVM visualization.

Burn Rate

The rate at which a program is consuming its budget over time.

Waterfall Chart

Shows how an initial value is affected by positive and negative changes; useful for variance analysis.

Trend Line

A line projected from historical data showing where the program is headed if current performance continues.

Data Normalization

Adjusting data to a common scale for meaningful comparison (e.g., cost per unit, not total cost).

DATA ANALYTICS FOR PROGRAM MANAGERS · LESSON 3

EVM Acronym Deep Dive: Every Term Defined and Explained

30 min

Master every EVM acronym (BCWS, BCWP, ACWP, CPI, SPI, EAC, ETC, VAC, TCPI) with clear definitions, formulas, and PM-level context.

Master every EVM acronym (BCWS, BCWP, ACWP, CPI, SPI, EAC, ETC, VAC, TCPI) with clear definitions, formulas, and PM-level context. Yes, it looks like alphabet soup. By the end of this lesson it will just look like a checklist.

The Six Formulas You Must Know Cold

$CV = BCWP - ACWP$. $SV = BCWP - BCWS$. $CPI = BCWP / ACWP$. $SPI = BCWP / BCWS$. $EAC = BAC / CPI$ (statistical). $VAC = BAC - EAC$. Everything else in EVM derives from these six.

Using EVM Indices Together: The 60-40 Rule and Beyond

Mid-career PMs know that no single EVM index tells the full story. The most robust EAC formula weights CPI and SPI: $EAC = BAC \div (0.2 \times SPI + 0.8 \times CPI)$. This formula reflects the empirical finding that cost performance has more predictive power than schedule performance for final cost outcomes. But use indices in combination: a program with CPI 0.95 and SPI 0.85 has both a cost problem and a schedule problem: the schedule pressure will likely drive overtime and accelerated work that further degrades CPI. It's the acquisition equivalent of sprinting the first mile of a marathon: it feels great for about ten minutes. Programs with $CPI \geq 1.00$ but $SPI < 0.85$ are common: they appear cost-efficient but the schedule pressure is a leading indicator of future cost overruns. Don't let a healthy CPI lull you into ignoring a deteriorating SPI.

The Cumulative CPI Stability Finding and What It Means for PMs

Research by Christle, Bhatt and others (the “Christle Findings”) established that once a program has expended 20% of its budget, the cumulative CPI rarely improves by more than 10% (confirming, with statistical rigor, what every experienced PM already suspected in their gut around month three). This has profound implications: a program at 20% complete with a cumulative CPI of 0.85 will, with very high probability, complete at a CPI no better than 0.85: meaning a 15%+ cost overrun is essentially locked in. Senior PMs use this finding to drive two actions: (1) early IBR (challenge cost realism before 20% is spent; (2) early OTB/EAC revision) if CPI is

deteriorating before the 20% mark, don't wait for it to stabilize. The longer you wait to acknowledge an overrun, the more the sunk cost fallacy traps the program. The cumulative CPI stability finding is the empirical basis for “fix it early or pay double.”

Key Terms

BAC

Budget at Completion: the total authorized budget for all work on the contract.

BCWS / PV

Budgeted Cost of Work Scheduled (Planned Value): what you planned to spend by today.

BCWP / EV

Budgeted Cost of Work Performed (Earned Value): the budget value of work actually completed.

ACWP / AC

Actual Cost of Work Performed: real dollars spent on completed work, from the accounting system.

CPI

Cost Performance Index = $BCWP / ACWP$. Below 1.0 means over budget.

SPI

Schedule Performance Index = $BCWP / BCWS$. Below 1.0 means behind schedule.

EAC

Estimate at Completion: projected total cost to finish all authorized work.

ETC

Estimate to Complete: projected cost to finish remaining work only.

VAC

Variance at Completion = $BAC - EAC$. Negative means projected overrun.

TCPI

To-Complete Performance Index: efficiency needed on remaining work to hit BAC or EAC.

Knowledge Check

A program has BCWP = \$800K, ACWP = \$1,000K, BCWS = \$900K. What is the CPI?

- A. 0.89
- ✓ B. 0.80
- C. 1.25
- D. 1.13

WHY

$CPI = BCWP / ACWP = \$800K / \$1,000K = 0.80$. For every dollar spent, only \$0.80 of value is earned: a 20% cost overrun rate.

Using the same data (BCWP=\$800K, ACWP=\$1,000K, BCWS=\$900K), what is the Schedule Variance?

- A. +\$100K (ahead)
- ✓ B. -\$100K (behind)
- C. -\$200K (behind)
- D. +\$200K (ahead)

WHY

$SV = BCWP - BCWS = \$800K - \$900K = -\$100K$. Negative SV means behind schedule: \$100K worth of planned work has not been completed.

What does TCPI > 1.10 indicate?

- A. Program is 10% ahead of budget
- ✓ B. Remaining work must be performed 10%+ more efficiently than history: likely unrealistic
- C. Program has a 10% cost overrun
- D. Program needs only 90% of remaining budget

WHY

TCPI of 1.10 means to stay within BAC, future work must be performed 10% more efficiently than the cumulative CPI. When TCPI significantly exceeds current CPI, the EAC should be revised upward.

The statistical EAC formula (BAC/CPI) is most reliable when:

- A. Contract is in its first month
- ✓ **B. After approximately 20% contract completion**
- C. Only when contractor provides bottom-up estimate
- D. During final 10% of contract

WHY

Research (RAND, GAO) consistently shows $EAC = BAC/CPI$ is among the most accurate predictors after approximately 20% completion: often more accurate than optimistic bottom-up estimates.

Why does Schedule Variance (SV) converge to zero at contract completion?

- A. All overruns are corrected at completion
- ✓ **B. BCWP and BCWS both equal BAC at completion, making $SV = 0$ regardless of lateness**
- C. Contractor reconciles variances before final payment
- D. Government accepts any date within 10% of plan

WHY

At contract completion all work is earned ($BCWP = BAC$) and all was planned ($BCWS = BAC$), so $SV = 0$ regardless of how late the program finished. Use Earned Schedule (ES) for late-program schedule analysis.

DATA ANALYTICS FOR PROGRAM MANAGERS · LESSON 4

Reading IPMR Reports: What Every PM Looks for in Each Format

22 min

Learn to read and analyze the six IPMR formats that contractors submit monthly on DoD programs, and understand what every government PM should look for in each one.

Learn to read and analyze the six IPMR formats that contractors submit monthly on DoD programs, and understand what every government PM should look for in each one. Six formats, one report, and about an afternoon of your life every month that you'll never get back.

What Is the IPMR and Why Does It Matter?

The Integrated Program Management Report (IPMR) is the primary contractually required data deliverable for EVM reporting on DoD contracts. It replaced the older Contract Performance Report (CPR) and Contract Funds Status Report (CFSR) through DI-MGMT-81861 (2012). If your contract exceeds \$20M cost-type or \$50M fixed-price, you will receive IPMR data from your contractor every month. This lesson teaches what each IPMR format contains and what government PMs look for.

Cross-Referencing IPMR Formats: The Monthly Analysis Workflow

Mid-career PMs don't just read IPMR formats sequentially: they cross-reference them. The highest-value cross-references: (1) Format 1 variance WBS elements → Format 5 narrative: does the narrative actually address the Format 1 problem areas, or does it address different elements? Inconsistency means the Format 5 was written independently of the data (more common than anyone likes to admit out loud in a program review); (2) Format 4 staffing gaps → Format 1 schedule variance: understaffed WBS elements should show negative SV: if they don't, the schedule logic is disconnected from staffing reality; (3) Format 5 corrective action commitments → next month's Format 1: are last month's corrective actions producing measurable improvement? If not, the corrective action was ineffective or was not actually executed. Build a monthly tracking sheet that captures the key cross-references from each IPMR submission.

IPMR as Legal Documentation: What Senior PMs Must Preserve

Senior PMs on large programs must understand that IPMR data is contractually required documentation and is discoverable in litigation. Format 5 narratives become exhibits in ASBCA cases: a contractor's written explanation that a variance is due to "government-caused delays" is the foundation of a future REA. The government PM's obligation: (1) preserve all IPMR submissions with receipt timestamps; (2) document

government responses in writing (if you accepted an inadequate Format 5 narrative without objection, you may have waived your right to challenge the root cause later; (3) track OTB and reprogramming history) baseline changes require formal government approval and must be documented with the approving official's name and date. Programs that lose IPMR archives lose their ability to reconstruct performance history in claims disputes.

Key Terms

IPMR

Integrated Program Management Report: the primary monthly EVM data deliverable from contractors on DoD programs, replacing the older CPR and CFSR.

Format 1

WBS Performance Summary: the most used IPMR format showing cost and schedule performance by WBS element.

Format 5

Explanations and Problem Analysis: written narratives from the contractor explaining root causes of variances and corrective actions.

Format 6

Integrated Master Schedule (IMS) in electronic format: shows network logic, critical path, and near-term milestones.

OTB

Over Target Baseline: a formal government-approved action that resets the Performance Measurement Baseline when the current baseline is no longer achievable.

Total Float

The amount of time a task can slip before it delays a downstream milestone or the contract completion. Negative total float means the task is already late.

VAC

Variance at Completion: the difference between BAC and EAC. Negative VAC means the program is projected to overrun budget.

Knowledge Check

Which IPMR format contains written narratives explaining root causes and corrective actions for significant variances?

A. Format 1, WBS Performance

B. Format 3, Baseline

✓ C. Format 5, Explanations and Problem Analysis

D. Format 6, IMS/Schedule

WHY

Format 5 contains the Variance Analysis Reports (VARs): written narratives that explain the root cause of significant cost and schedule variances and describe corrective actions. It is the most important format for understanding WHY performance is off track.

A government PM reviewing Format 3 notices the PMB was retroactively changed to reduce a prior period's negative SV. This most likely indicates:

A. Normal baseline maintenance per EVMS guidelines

✓ B. A potential EVMS compliance violation: retroactive replanning to conceal schedule variances

C. An approved change order requiring replanning

D. A subcontractor claiming additional scope

WHY

Retroactive changes to the PMB to reduce prior period variances are a serious EVMS compliance concern. ANSI 748 and DoDI 5000.02 prohibit retroactive baseline revisions except in very specific circumstances. This is classic “rubber baseline” gaming behavior.

Format 4 shows actual headcount is 30% below planned for the critical design phase. What should you expect in Format 1?

A. Positive CPI due to lower labor costs

✓ B. Negative SV because insufficient staff means less work is being accomplished

C. Positive SV because fewer people compresses the schedule

D. No impact: staffing and EVM performance are independent

WHY

Understaffing is a direct cause of negative Schedule Variance. Fewer people completing work means BCWP accumulates slower than BCWS, creating negative SV. Cross-referencing Format 4 staffing gaps with Format 1 schedule variances is a fundamental IPMR analysis technique.

In Format 6 (IMS), a task shows “-15 days total float.” What does this mean?

A. The task has 15 days of scheduling flexibility

✓ B. The task is already 15 days late relative to its constraint and will delay the program end date

C. The task was completed 15 days early

D. The task needs 15 additional resources

WHY

Negative total float means the task is already past its planned completion date by that amount. Without recovery actions, negative float propagates through the network and delays all successor tasks including the contract completion milestone.

What is an Over Target Baseline (OTB) and when is it used?

A. An informal replanning action to reset performance metrics

✓ B. A formal government-approved reprogramming action that resets the PMB when the current baseline is no longer achievable, requiring a new IBR

C. A penalty mechanism for contractors who exceed the BAC

D. A supplemental funding authorization for overrun programs

WHY

An OTB is a formal, government-approved action that resets the Performance Measurement Baseline when the current baseline is so unrealistic that EVM measurements have lost informational value. It requires SSA/PEO approval, a revised EAC, and a new Integrated Baseline Review (IBR). OTB does not change the contract price.